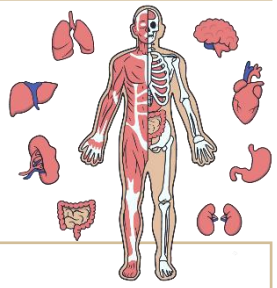
Module | Anatomy

Cours nº01 : Introduction to human anatomy

Academic Year : **1ere année médecine 2025-2026**

Date : **06/05/2025**

Prof : **Dr M.ZENNIR**



Course Plan :

1. **Definition**
2. **Branches of anatomy**
3. **Anatomical nomenclature**
4. **Organization of the human body**
5. **Regions of the human body**

1. Definition :

-Human anatomy is a morphological and descriptive science which studies the forms and structures of the human body.
-the word anatomy is derived from the greek word **anat ome** which means to cut up
-the latin equivalent of this is dissecare from which the word dissection is derived .in the laboratory, anatomy is studied by **dissection** (sect = cut, dis = apart).
-of all the basic sciences offred to students of medicine dentistry and allied health sciences there is none more directly related to their professional
-practice than anatomy and its application :surgery, medical ,imagery ……

2. Branches of anatomy :

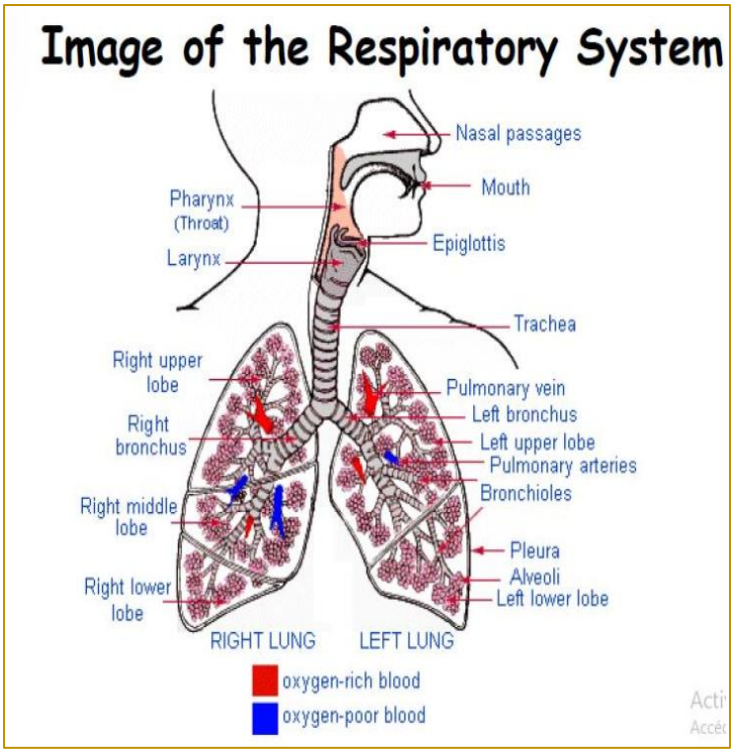
Two major types of anatomy are :

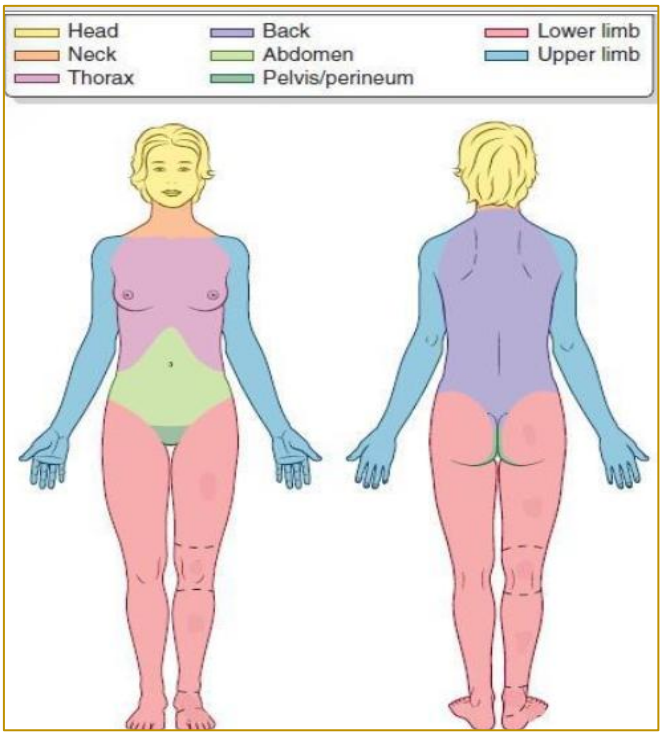
- **Gross anatomy :**

-The study of the parts and structures of the human body that can be seen with the naked eye and without a microscope
-by systems is the study of organ system (like the respiratory system)
-by regions considers such as the trunk, upper member, or lower member.

- **Microscopic anatomy :**

-The study of the parts and structures of the human body that can NOT be seen with the naked eye and only seen with the use of a microscope
➔ the study of tissues (Histology)
➔ the study of cells (Cytology)

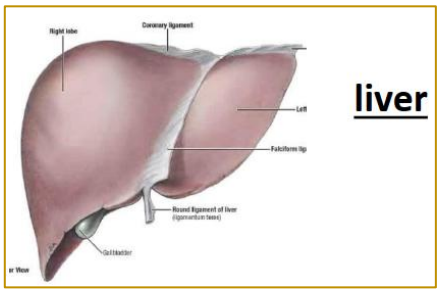




Regions of the human body

According to the point of interest, there are several subtypes of anatomy:

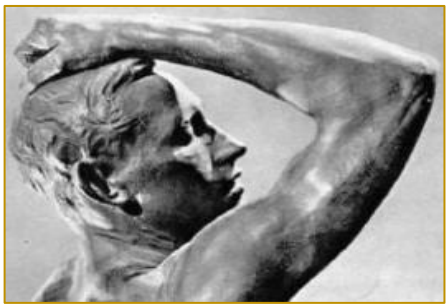
-Descriptive anatomy describes the structure of body parts, describing their location, orientation, shape, dimensions, weight, colour, texture, internal structure, etc. **it is the basic study in anatomy**

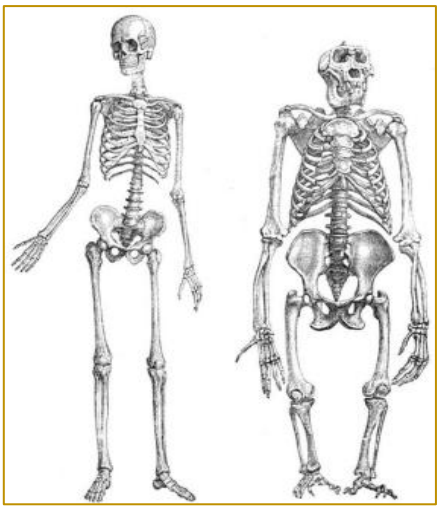


-Regional anatomy describes body by defined body parts and deals with the relationships between various
organs of the same anatomical region and their projections onto the skeleton and body surface. **It is the basic study of surgery**



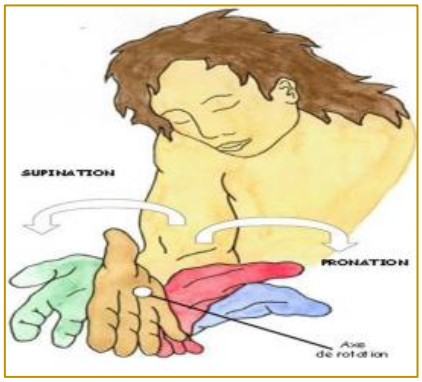
-Superficial or palpatory anatomy is the study of the external forms and reliefs of the body.
It is the basis for clinical observation by **palpation**, and for artistic anatomy.(sculptor ;painter)





-Comparative anatomy studies the evolution of the species The study of **similarities** and **differences** In the anatomy of two species .

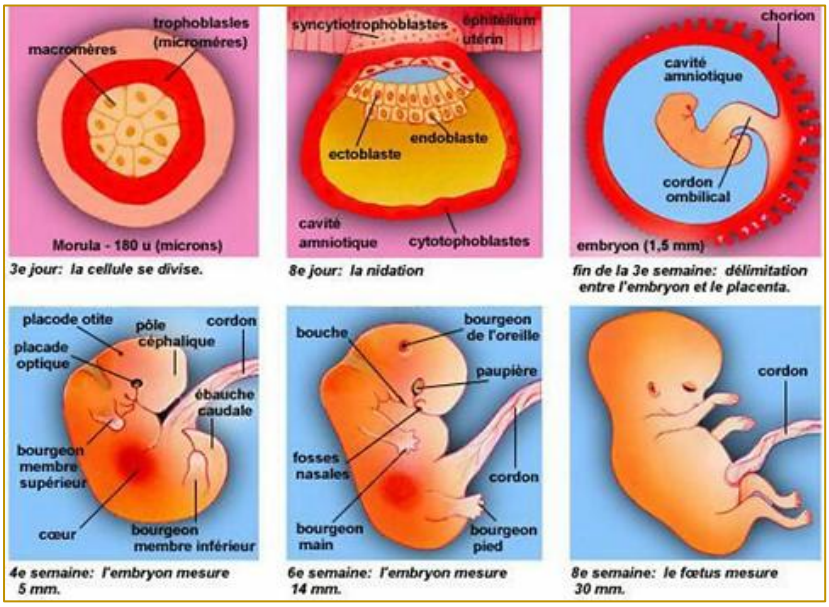
-Functional anatomy Studies the **relationships** between the morphology and the functions of organs.

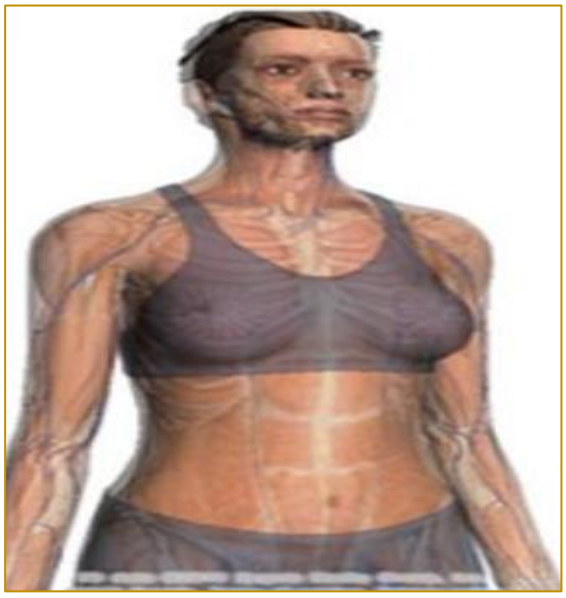


-Radiological anatomy Study of morphology using **imaging techniques** such as X-rays, CT (Computed Tomography), MRI (Magnetic Resonance Imaging), and scintigraphy



-Developmental anatomy studies of the morphological transformations of the organism from fertilization to adulthood.





3. Anatomical nomenclature :

-Like all sciences anatomy has its own language entended to facilitate communication and Understanding .
-For a universal consensus on anatomical language in 1955 at the 6th federal congress of anatomy in paris;it was decided to adopt a latin and international terminology or nomina anatomica **(NA)**.

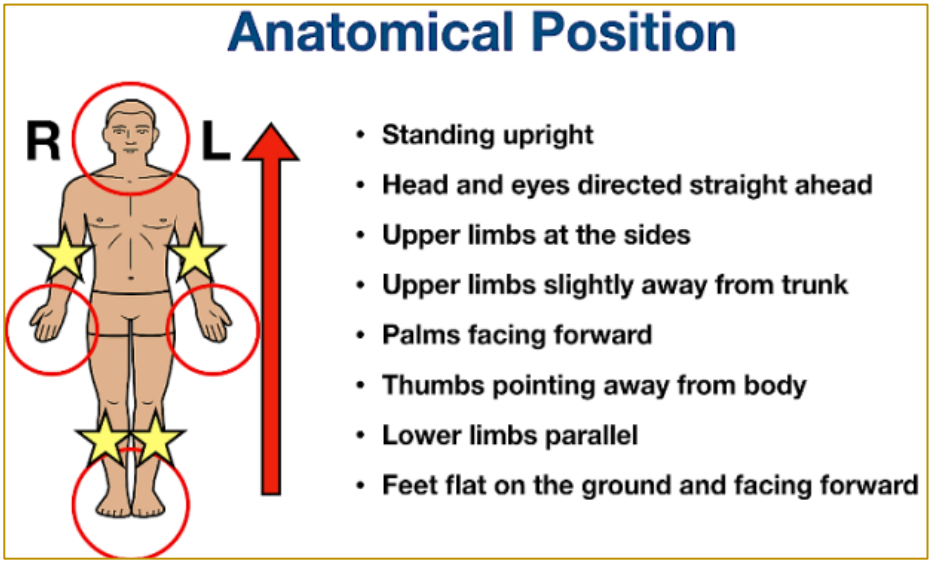
Anatomical terms constitue the basis of medical language.

-This nomenclature refers to a well-defined position of the human body, in which three axes are distinguished. These axes allow the definition of three planes, along with reference points that indicate the direction of movements.

- **The anatomical position :**

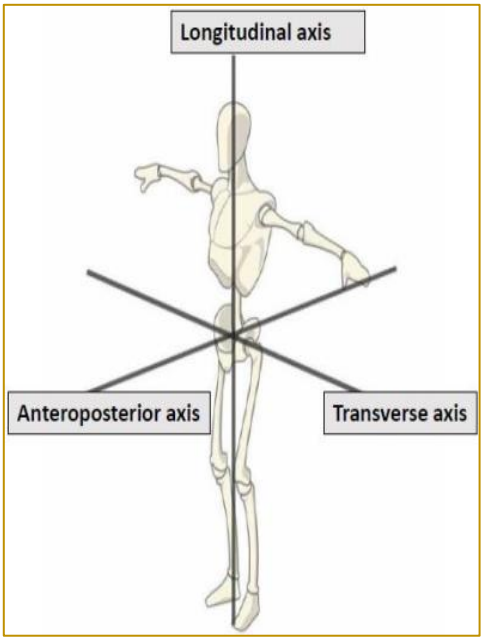
For each description, a marker or origin position is obligatory ;The human reference position, known as the standard anatomical position, Is defined as
-a person standing erect,
-with upper limbs extended by the side of the trunk with the palms facing forward (thumbs outward),
-lower limbs extended with the feet parallel and together,

-and the head facing forward, the gaze directed toward the horizon.



- **Reference anatomical axes :**

-Longitudinal axis of the body : It is an imaginary straight line connecting the top of the head to the foot.



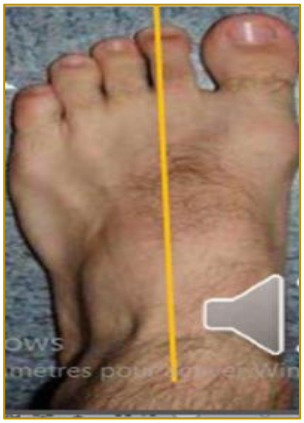
-Anteroposterior axis : It is an imaginary line that passes through the body from front to back.

-Transverse axis : An imaginary line that passes through the body from left to right.

-Hand axis : Longitudinal passing through the third finger

-Foot axis : the horizontal axis passes through the second toe





- **The reference planes :**

Three reference groups of planes they are perpendicular to each other.

-Sagittal plane :Is vertical plane that pass through the body from front to back.it divides the body into right and left parts (2 symetric halves)

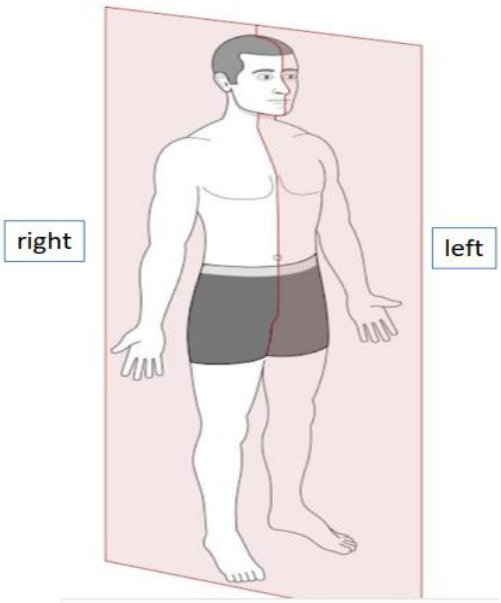
-The paramedian plane : Is plane running parallel to the median plane ;divides the body into Medial, lateral

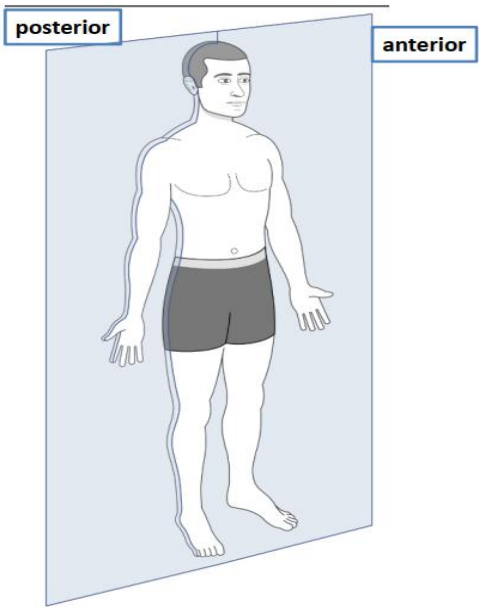
-Frontal (coronal) plane :

-is vertical plane that pass through the body from side to side.

-that is parallel to the forehead. And perpendicular to the sagittal plane

-Divides the body into. Anterior and posterior parts

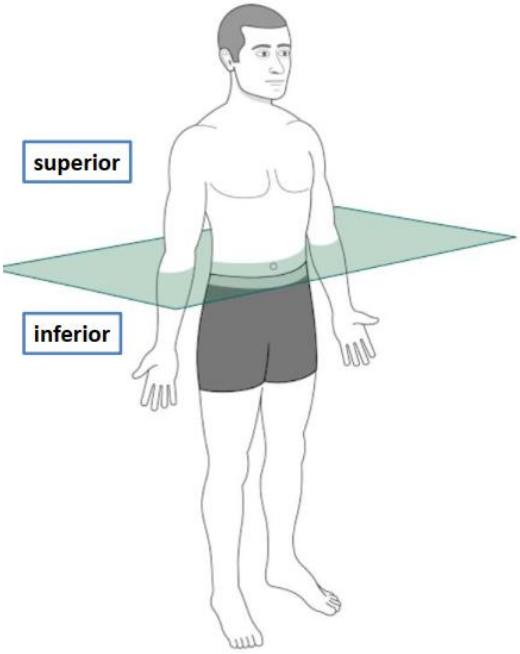


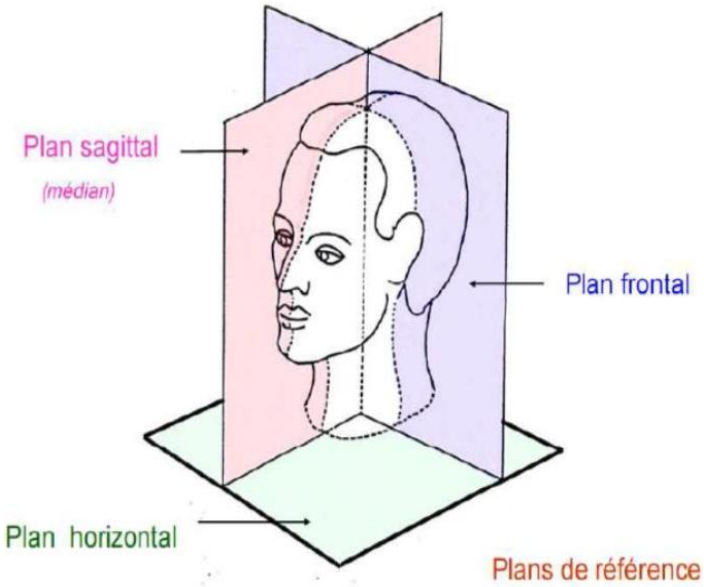


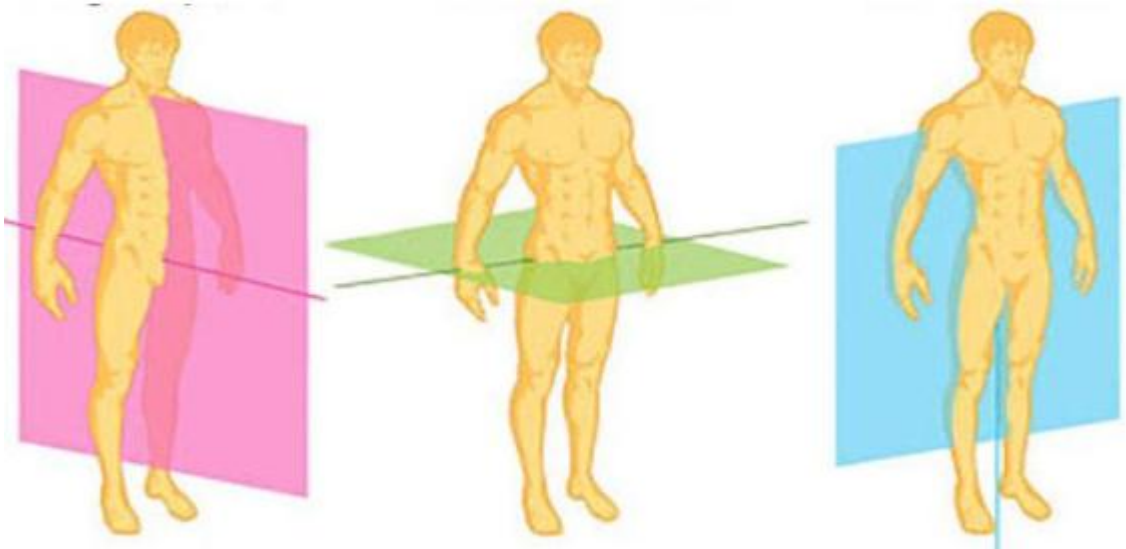
-The transverse (horizontal) :

-Plane is **parallel** to the floor and **perpendicular** to both the sagittal and frontal planes.

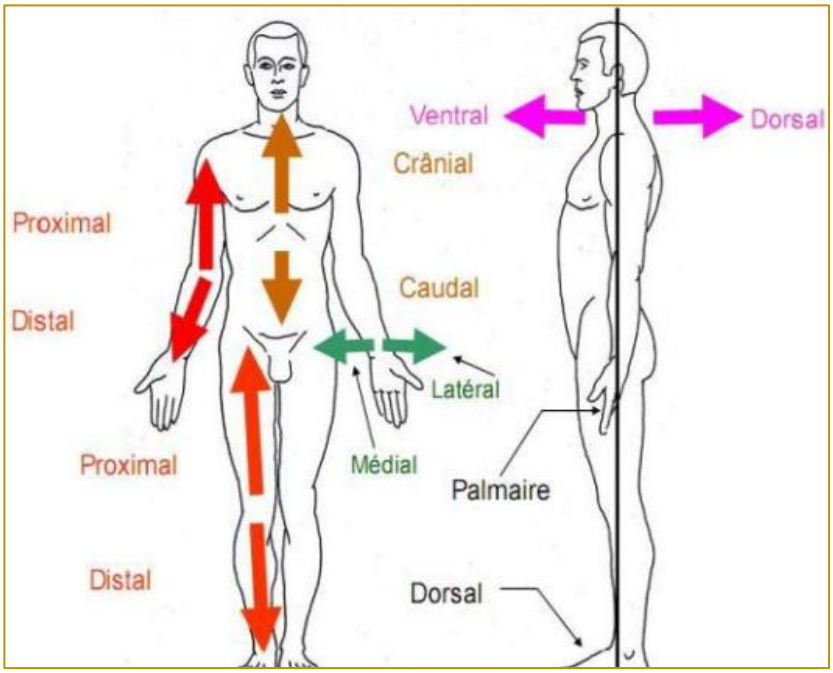
-divides the body into Superior (upper) inferior(lower)parts







-Terms of location and direction :Anatomical directional terms (of relation) are used to describe the position of each body structure and relate it to the positions of other structures.



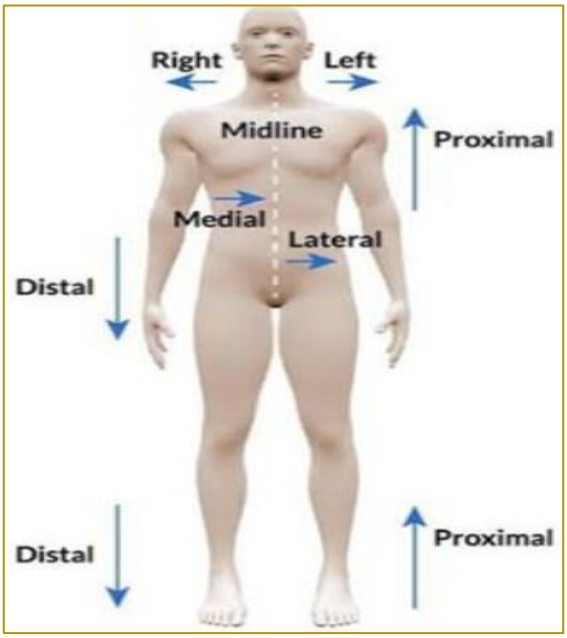
- ⇨ *Proximal :* is nearer to the point of attachment to the body (on the limbs).

eg. the arm is proximal to the hand.

- ⇨ *Distal :* is further away from the point of attachment to the body (on the limbs).

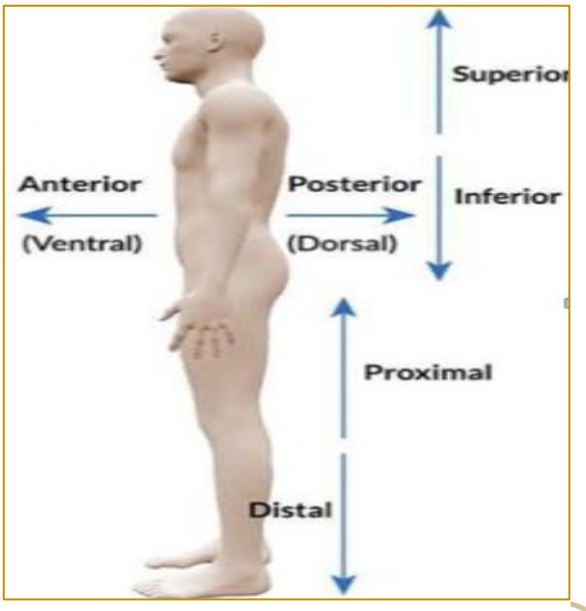
e.g. the hand is distal to the arm.

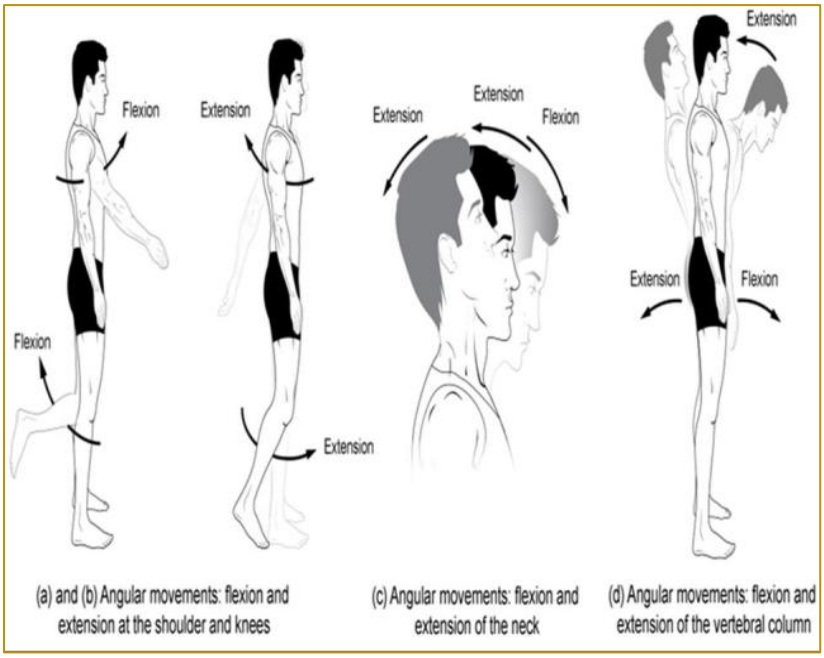
- ⇨ *Medial or internal:* is nearer the midline of the body.
- ⇨ *Lateral or external :* Is far from the midline.
- ⇨ **Superficial** is nearer the skin.
- ⇨ **Deep** is away from the skin.
- ⇨ **Anterior (ventral) :** is nearer the front of the body. In the hand it is replaced by **palmar**.
- ⇨ **Posterior** (**dorsal) :** nearer the back.
- ⇨ **Superior** (**cephalic** or **cranial).** is nearer the head
- ⇨ **Inferior (caudal)** is nearer the feet. in the feet it is replaced by and by **plantar**.
- ⇨ **Homo-lateral:** same side
- ⇨ **Contra-lateral:**opposite sides

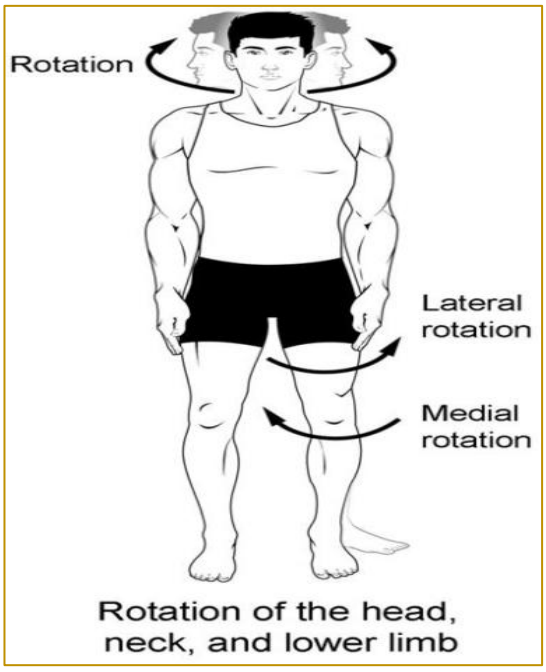


-Terms of Movements : Movements occur around joints and are referred to them ;this movments are done in a plan by relation to an axis

- ⇨ *Flexion & Extension***:** to flex a joint is to fold it (i.e. decrease the angle) and to extend a joint is to unfold it (i.e. increase the angle). This movement is done in the saggittal Plane along the transversal axis
- ⇨ *Rotation:* this movement occurs around a longitudinal axis either towards the midline (Medial or Internal Rotation) or away from the midline (Lateral or External Rotation). This movement is done in the transversal Plane along the vertical axis

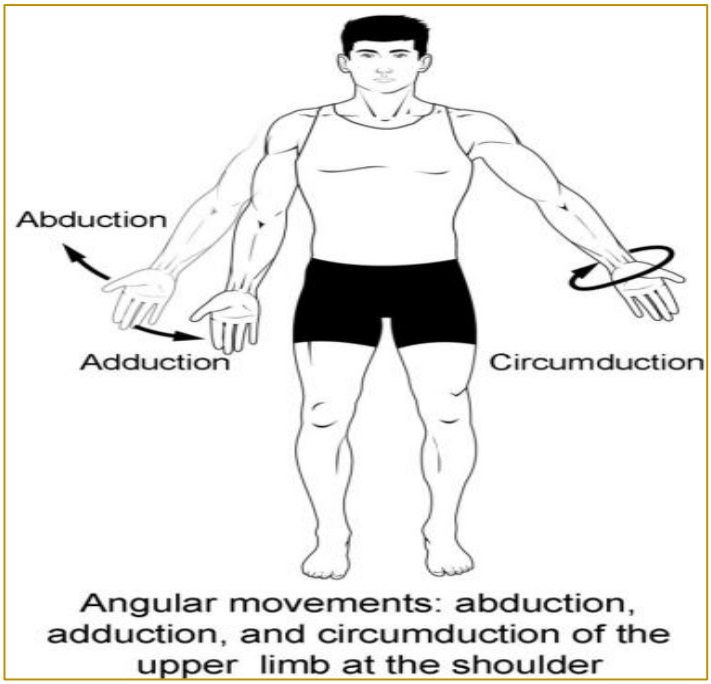






⇨ *Abduction and adduction:* to abduct is to move away from the median plane (increasing the angle) and to adduct is to move towards the median plane (decreasing the angle).
This movement is done in the frontal plane along the anteroposterieor axis

Circumduction: is a conical rotatory movement creating an arc around a longitudinal axis and represents the combination of flexion, abduction, extension & adduction



4. Organization of the human body :

the human body is organized in several levels, from the simplest to the most complex. .from the cells to the total organism.

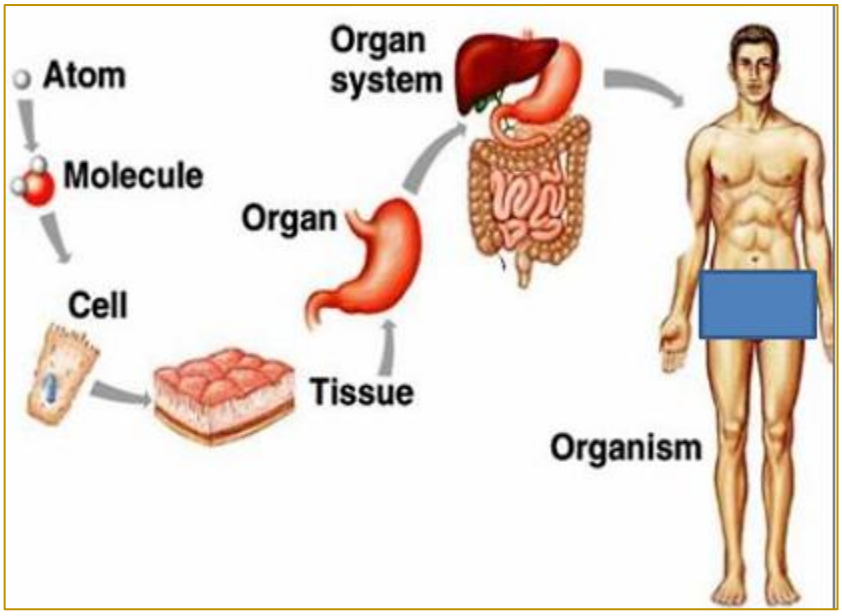
-Cells are the smallest living unit of body construction there are four types of cells epithelial, connective, muscular, and nervous.

-tissue is a grouping of like cells working together. Cells with the same specialisation form tissue. It performing a similar function

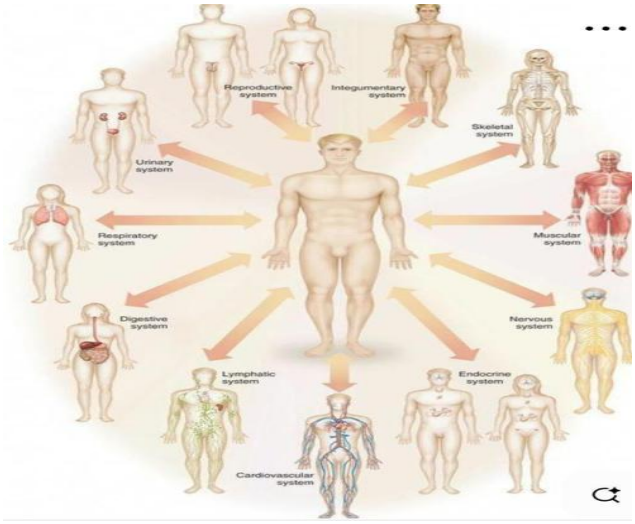
-Examples are muscle tissue ;nervous tissue.

-organ is a structure composed of several different tissues. that performs a specific function
Examples (stomach, kidney, bone).

-Organ systems are groups of organs which together perform an overall function.
-Examples are the respiratory system and the digestive system.



The 11 human body systems are as follows:
1. Integumentary system (skin)
2. skeletekel system (Bones)
3. Muscular system (muscles)
4. The nervous system.
5. The endocrine system.
6. The cardiovascular system
7. The immune system
8. The digestive system
9. The respiratory system
10. The urinary system
11. The reproductive system



- **Organisation of the human body in apparatuses :**

-The apparatuses of relational life: it included all of the apparatuses that :
-put the human body in contact with its environment
-its assures the maintenance of posture and movements
-its provides the reception of information from the environment, its integration and the corresponding reactions
Muscular system (muscles),skeletekel system(Bones),The nervous system.the sense organs
-The apparatus of vegetative life : which ensure the self-regulation of the human body and which are autonomously managed
The endocrine system. The digestive system. The respiratory system.
The cardiovascular system. The urinary system. The immune system.
-The apparatus of sexual reproduction :which enables the continuation of the human species
the male genital system;
the female genital system

5. REGIONS OF THE HUMAN BODY :

-Head The head is divided into the cranium part and the face.

-**Neck**. Is a passage way for visceral; vascular and nervous elements to or from the head

-**Trunk**.(torso) is divided in three parts thorax, abdomen and pelvis

-upper member includes

-a shoulder,

-arm,

-elbow

-forearm,

-wrist,

-hand.

-Fingers

lower member includes

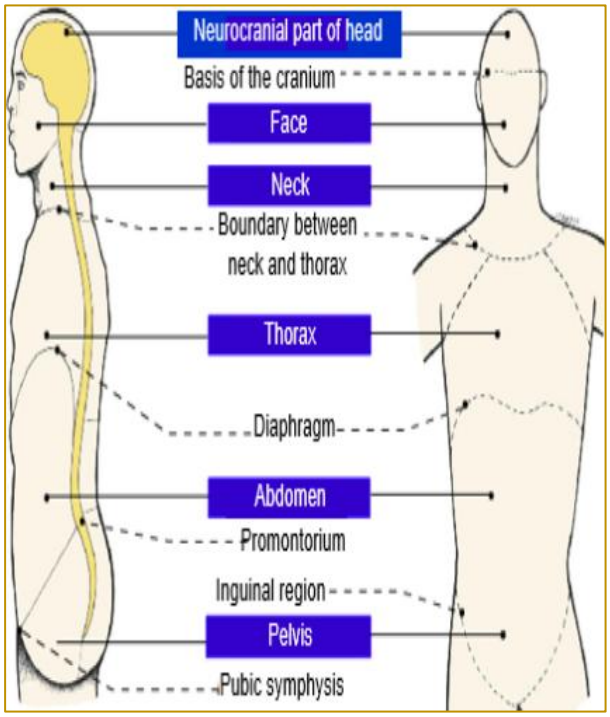
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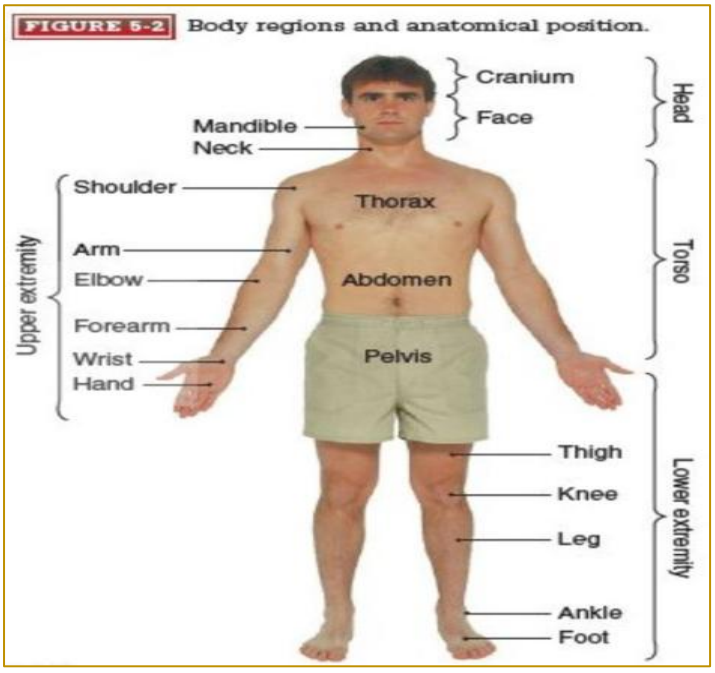
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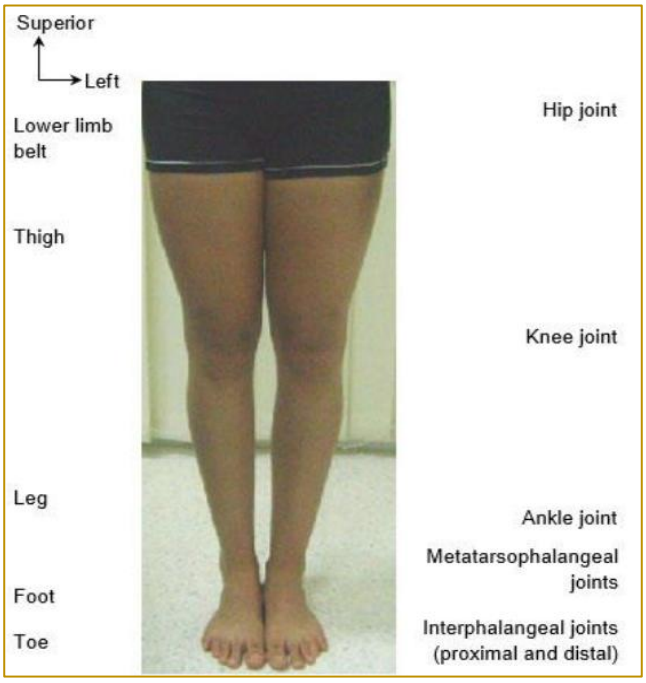
-leg,

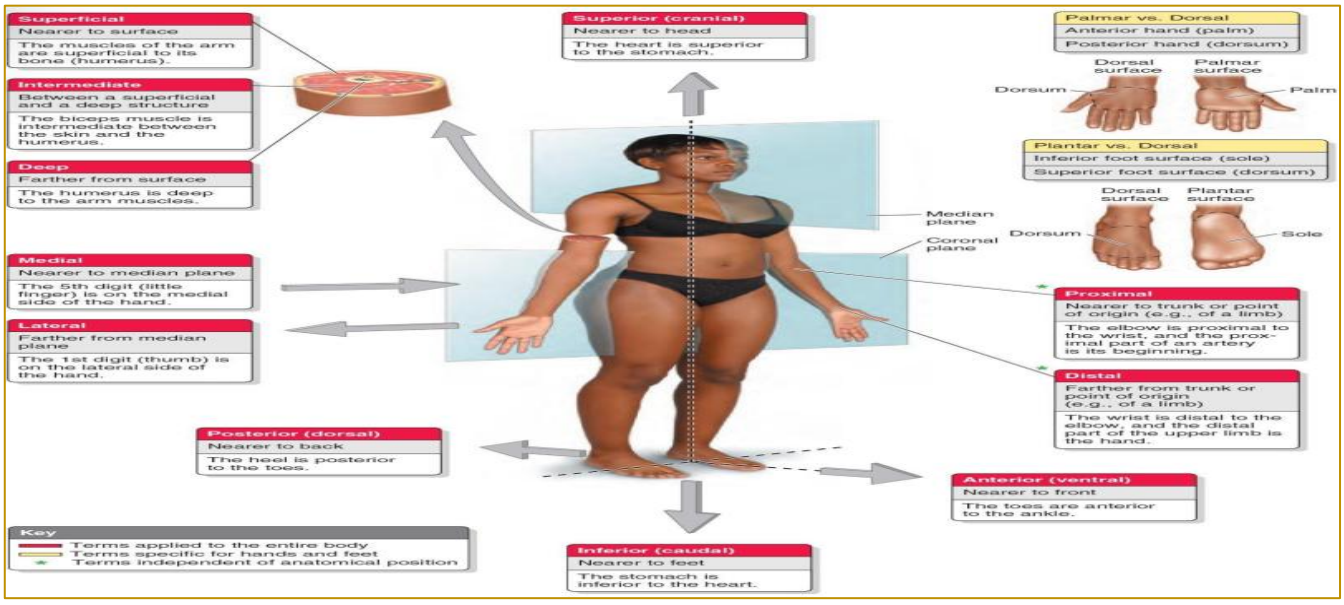
-ankle,

-foot.and toes









Notes

Nombre de couches